

Epigenetic Quantile Age (EQuA) Clocks: A New Paradigm for Evaluating Biological Aging Acceleration

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EQuA GitHub repository

01 WHY A SINGLE EPIGENETIC AGE ESTIMATE IS NOT ENOUGH

Biological Age: Background and Motivation

Chronological age (CA). Time elapsed since birth.

Biological age (BA). A latent physiological aging state that reflects health status and functional decline.

Fact. People of the same chronological age may differ substantially in biological aging.

BA < CA
Biologically younger

BA > CA
Biologically older

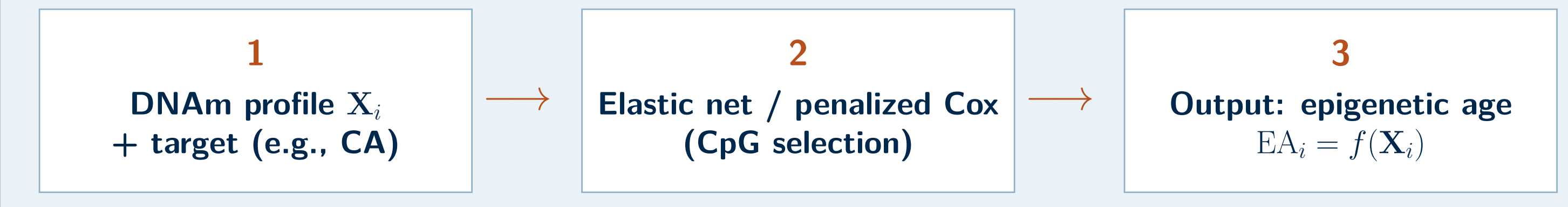
Scientific Questions

- How can biological aging be **quantified**?
- How can aging **heterogeneity** be characterized?
- What factors **accelerate or decelerate** biological aging?

Epigenetic Clocks: From DNA Methylation to an Age Estimate

DNA methylation data. Illumina 450K and EPIC arrays measure approximately 450,000 and 850,000 CpG sites per person, producing an ultra-high-dimensional methylation profile $\mathbf{X}_i = (X_{i1}, \dots, X_{ip})^T \in [0, 1]^p$.

How an epigenetic clock is trained



Representative Clocks

Clock	Reference	Tissue	CpGs	Training target	Method
Horvath	Horvath (2013)	Multi-tissue	353	Chronological age	Elastic net
Hannum	Hannum et al. (2013)	Blood	71	Chronological age	Elastic net
PhenoAge	Levine et al. (2018)	Blood	513	Phenotypic age	Cox → Elastic net
GrimAge	Lu et al. (2019)	Blood	1,030	Mortality risk	Elastic net → Cox
DNAmTL	Lu et al. (2019)	Blood	140	Telomere length	Elastic net
PedBE	McEwen et al. (2020)	Buccal	94	Pediatric age	Elastic net
Wu	Wu et al. (2019)	Pediatric blood	111	Pediatric age	Elastic net

Wu clock: the pediatric blood clock used in the ELEMENT application.

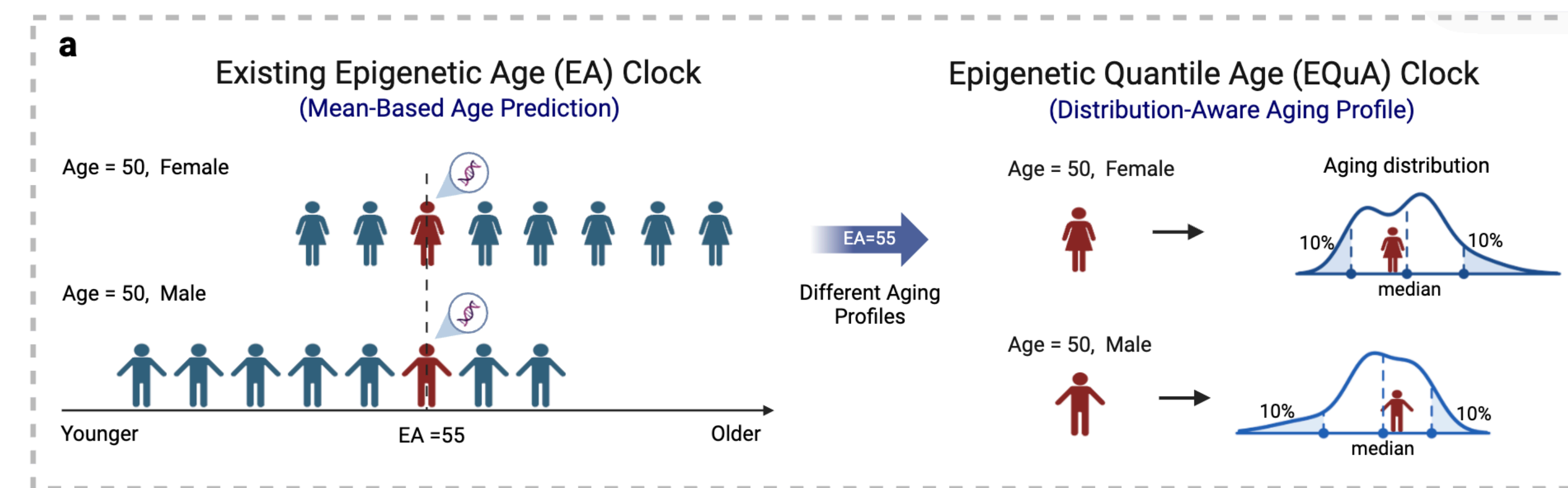
A Key Limitation of Existing Clocks

Regardless of their training targets, existing epigenetic clocks **compress** high-dimensional DNA methylation profiles into a **single scalar output**. Consequently, a single number cannot reveal whether aging differences extend across the age distribution, are concentrated in its lower or upper tail, or vary by percentile.

Beyond “How Old?”

Can the same epigenetic age correspond to different aging profiles?

EQuA transforms a single epigenetic age estimate into a percentile-resolved aging profile.



Proposed EQuA

- A Richer Aging Representation**
Characterizes profile location, spread, percentile-dependent separation, ordering reversals, and profile crossing.
- Distributional Transfer of Existing Clocks**
Extends a fixed scalar clock into a percentile-resolved output without accessing the original training data or refitting its CpG coefficients.
- Profile-Resolved Association Analysis**
Identifies where along the aging profile an association occurs.

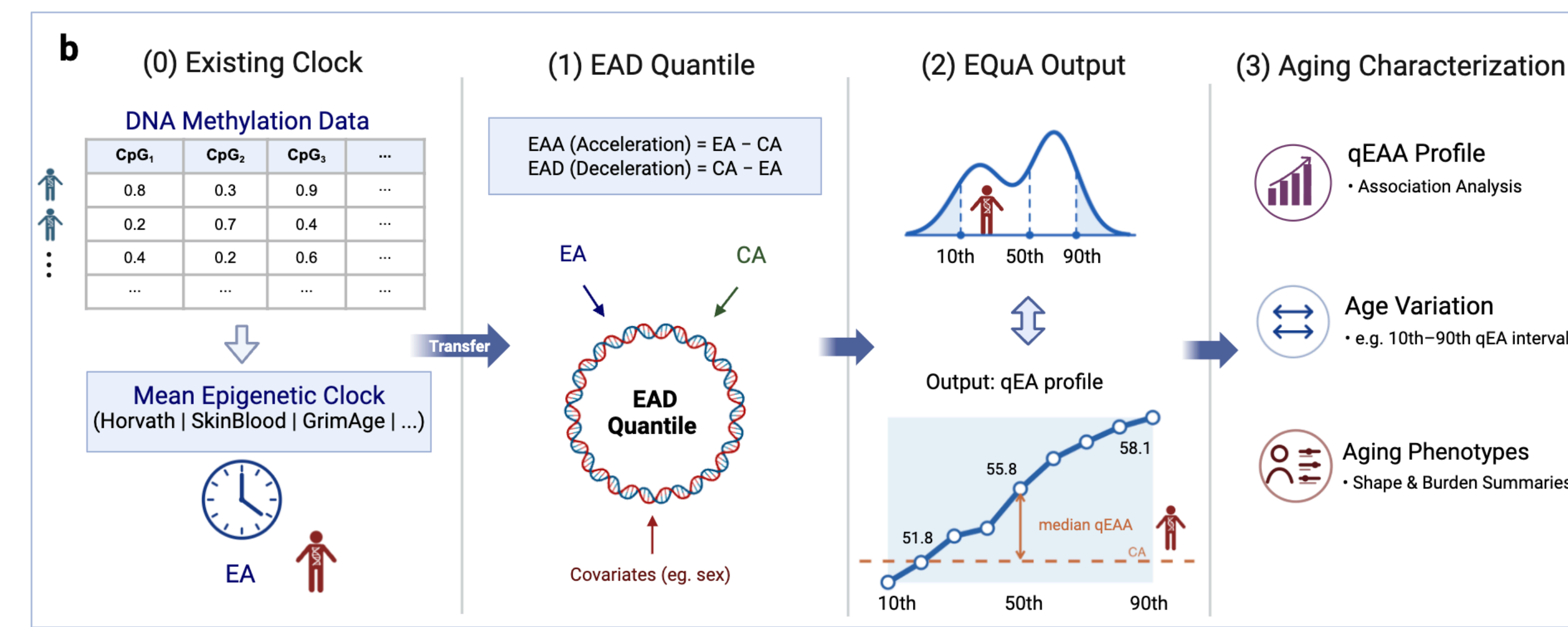
02 EQuA: FROM AN EXISTING CLOCK TO AN AGING PROFILE

EQuA Framework

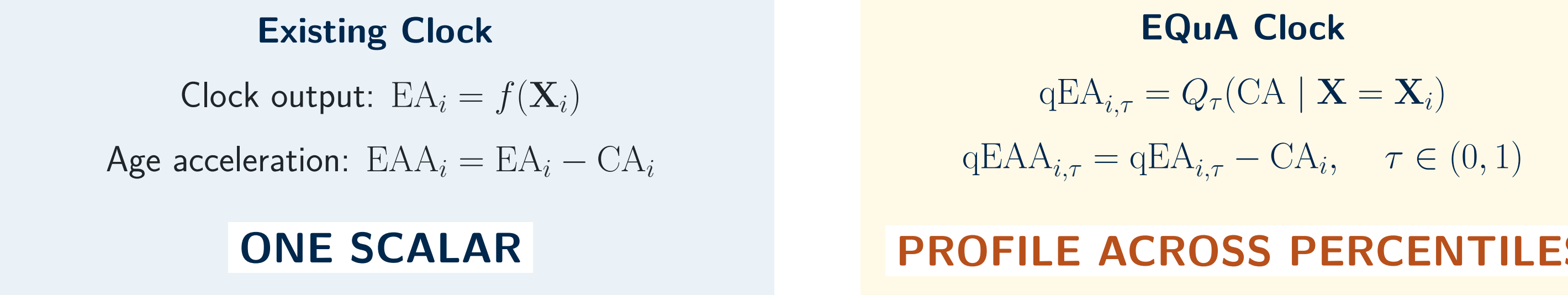
For a person with DNAm profile $\mathbf{x}$, the EQuA clock defines the τ th **quantile epigenetic age** as:

$$Q_\tau(\text{CA} | \mathbf{X} = \mathbf{x}) = \text{EA}(\mathbf{x}) + Q_\tau(\text{EAD} | \mathbf{X} = \mathbf{x}), \quad \tau \in (0, 1),$$

where $\text{EAD} = \text{CA} - \text{EA}(\mathbf{X})$ is epigenetic age deceleration, the sign-reversed epigenetic age acceleration.



From a Scalar Estimate to a Percentile-Resolved Profile



Three EQuA Formulations

Direct conditioning on the high-dimensional DNAm profile $\mathbf{X}$ would require fitting a new high-dimensional quantile model. EQuA instead approximates $Q_\tau(\text{EAD} | \mathbf{X})$ using three low-dimensional conditioning sets while keeping the source clock fixed.

1. Po-EQuA

Uses the marginal EAD distribution, assuming that residual age variation is homogeneous across DNAm profiles.

$$Q_\tau(\text{EAD})$$

2. End-EQuA

Uses the fixed-clock EA as a one-dimensional summary, assuming that it captures the relevant residual quantile structure.

$$Q_\tau\{\text{EAD} | \text{EA}(\mathbf{X})\}$$

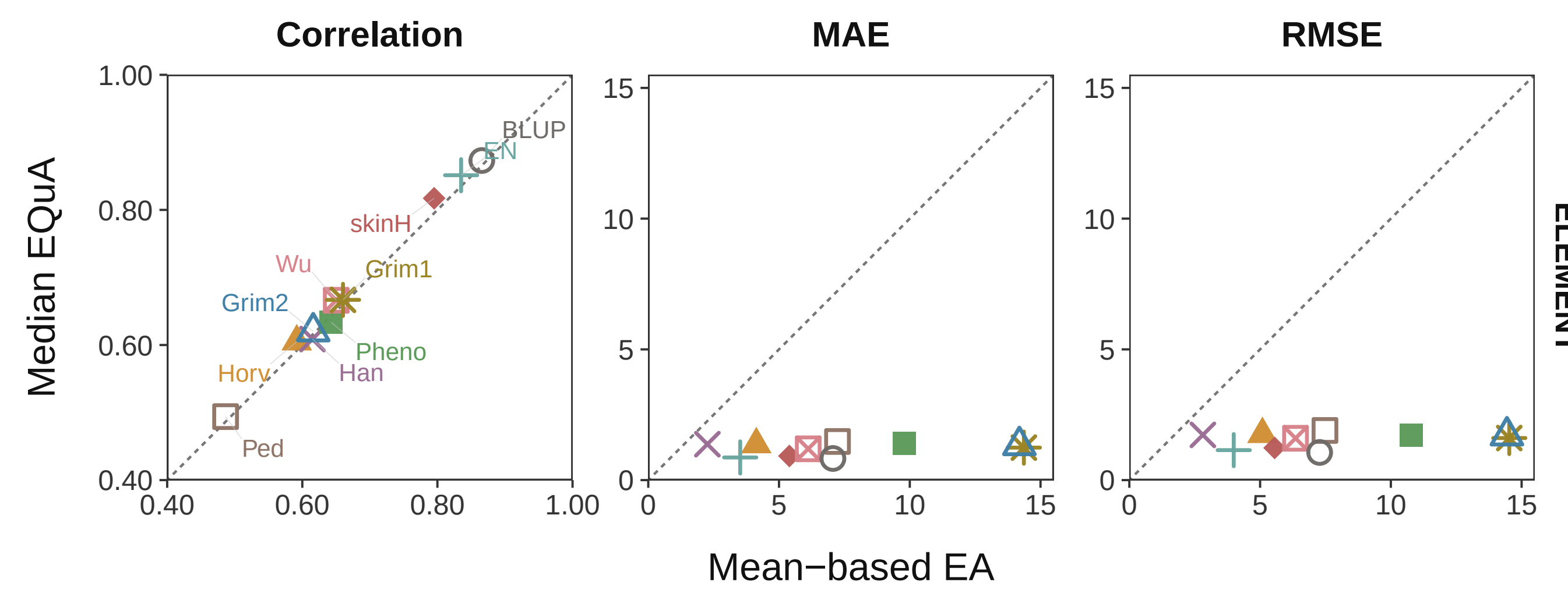
3. Ex-EQuA

Adds exogenous auxiliary variables $\mathbf{Z}$ (e.g., batch, sex) to capture relevant residual structure not retained by the fixed-clock EA.

$$Q_\tau\{\text{EAD} | \text{EA}(\mathbf{X}), \mathbf{Z}\}$$

Cross-Fitted Performance in ELEMENT Cohort

- ELEMENT** is a Mexico City birth cohort focused on environmental exposures and child health (Perng et al., 2019).
- We analyzed **526 participants aged 9.8–18.1 years** with blood DNA methylation data.
- Across **10 epigenetic clocks**, original EA was compared with cross-fitted **median EQuA**.



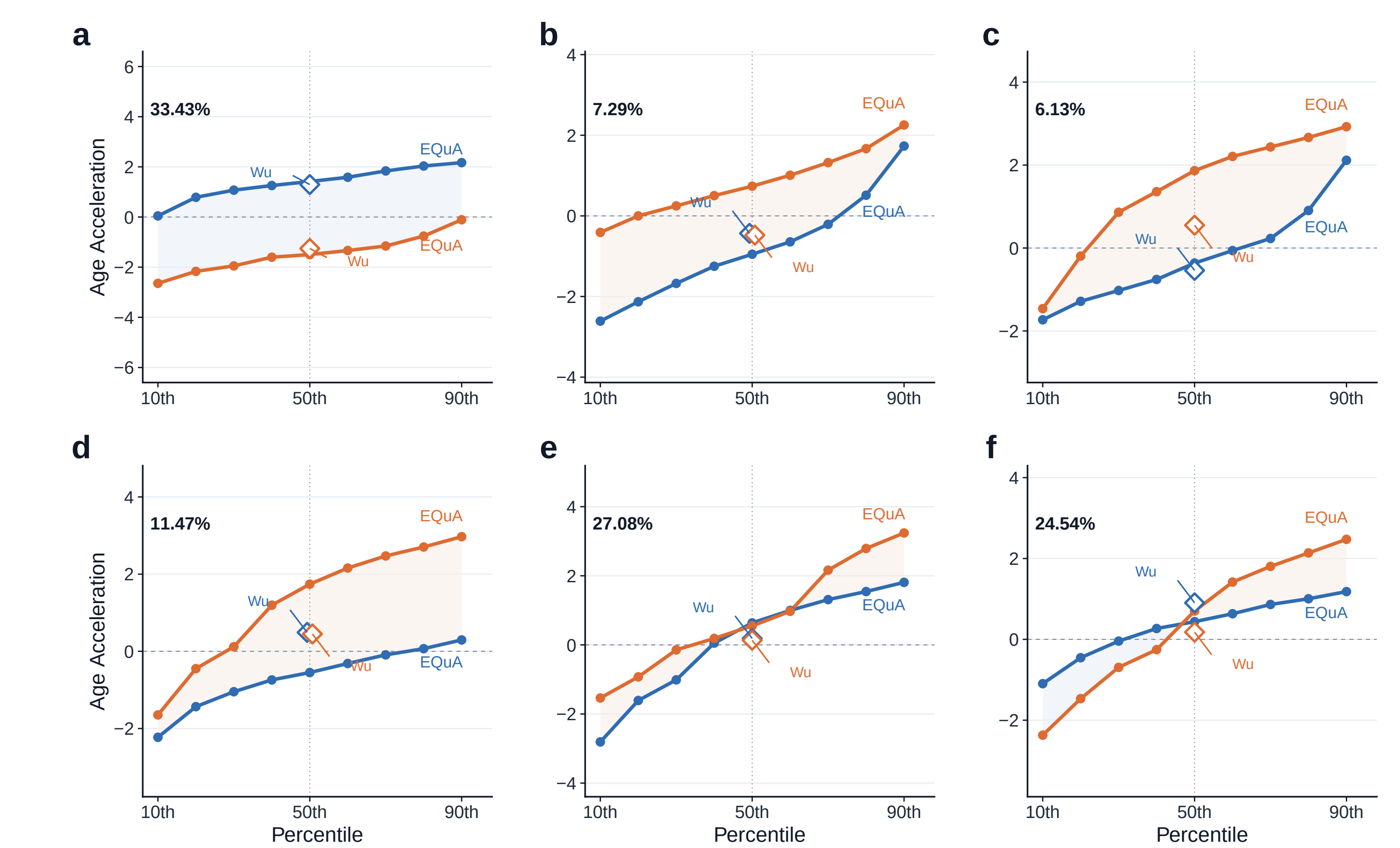
- Correlation:** above the diagonal favors EQuA; **MAE/RMSE:** below the diagonal favors EQuA.
- EQuA generally reduced age-prediction errors while preserving correlation with chronological age.**

Δ CORRELATION +0.0075 correlation preserved	Δ MAE −5.34 y lower prediction error	Δ RMSE −5.17 y lower prediction error
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03 WHAT THE EQuA AGING PROFILE REVEALS

Different Aging Profiles Between Two People in ELEMENT

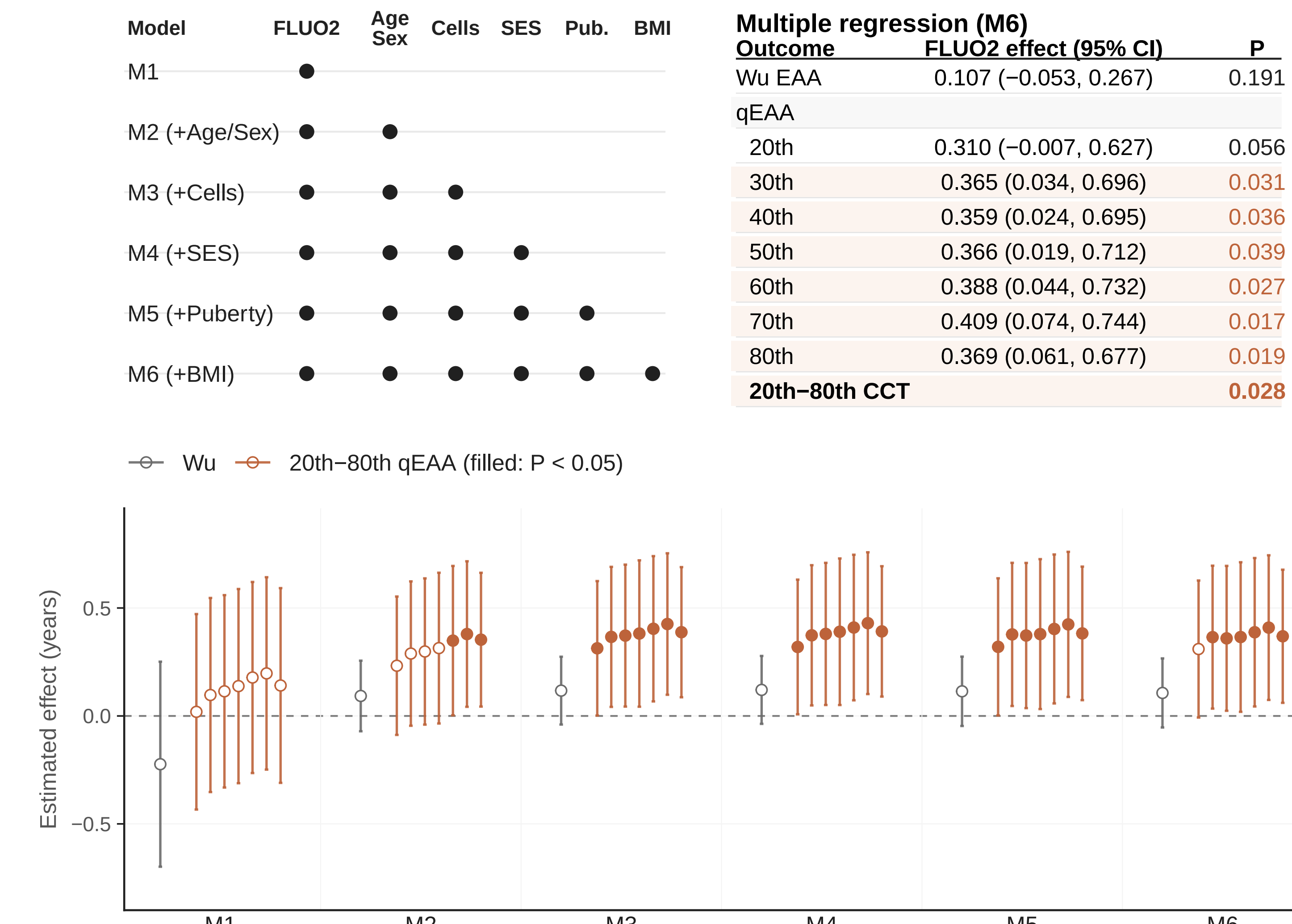
- We illustrate how age-acceleration patterns can differ **between two people**.
- Percentages indicate the proportion of **pairs of people exhibiting each pattern**.
- Wu EAA captures broad between-person ordering in some pairs, but cannot represent differences in qEAA profile **level, spread, separation, or relative ordering** across percentiles.



a, Wu EAA and qEAA profiles show concordant between-person ordering. *b–d*, Similar scalar EAA can mask differences in profile level, percentile-dependent separation, and profile width. *e*, Wu EAA ordering disagrees with the ordering across most qEAA percentiles. *f*, qEAA profiles cross, so relative ordering changes across percentiles.

PAH Exposure and Epigenetic Aging in ELEMENT

- FLUO2** is a urinary biomarker of exposure to fluorene, a PAH commonly found in traffic and industrial emissions, tobacco smoke, emissions from indoor fuel burning, and grilled foods.
- Outcomes:** Wu EAA and EQuA qEAA profiles.
- Model:** FLUO2, age, sex, blood-cell composition, socioeconomic status, pubertal stage, and BMI.
- Question:** Can EQuA profiles reveal a FLUO2-associated aging pattern that is not apparent with Wu EAA?



Higher FLUO2 → **Wu EAA:** no clear association ($\beta = 0.107$, $P = 0.191$).
Higher FLUO2 → **EQuA profile:** broadly higher qEAA across percentiles (CCT $P = 0.028$).
EQuA reveals where an exposure association occurs across the aging profile.